

Optimal Sizing of Battery Energy Storage Systems Considering Degradation Effect for Operating and Electricity Cost Minimization in Microgrids

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A microgrid is a low-voltage distribution network designed to provide power for small-scale and isolated communities consisting of distributed generation and energy storage systems. In order to achieve reliable power and proper energy utilization, energy storage systems play a vital role in microgrids by storing energy during off peak hours and discharges energy during peak hours. One of the major issues in the isolated microgrids with intermittent nature of distributed generations is the balance of energy demand. This can be achieved by appending renewable energy sources with suitable battery energy storage systems (BESS), to provide the reserve support in meeting the load demand. Battery degradation effect plays a major role in analyzing the performance of BESS lifetime. Battery degradation effect relates the capacity reduction of energy of BESS that is to be delivered to meet the load demand. Therefore, microgrid systems with BESS considering degradation effect should be optimized in such a way as to obtain minimum operating cost while ensuring minimum electricity cost to customers. Simulations of hourly battery discharge rates and simulation of actual discharge rates with obtained simulated rates need to be calculated to determine the degradation effect. The battery degradation cost and lifetime can be calculated correspondingly to minimize the objectives. Correspondingly, herein, an optimization algorithm for microgrid operating cost and customer electricity cost minimization for 24 h time horizon while considering BESS degradation effect by determining kWh and MWh throughput is presented. Particle swarm optimization (PSO), accelerated particle swarm optimization (APSO), Jaya optimization (JAYA) technique, and linear programming interior point algorithm (LP-IP) are applied to determine the optimal operating cost and electricity cost by simulating BESS degradation parameters. Results obtained using PSO, APSO, JAYA are compared with LP-IP solver-based algorithm for BESS lifetime, BESS degradation cost, microgrid operating cost, and customer electricity cost for 24 h.

1. Introduction

Microgrids with distributed energy resources (DERs) have become one of the promising solutions of future smart grids. However, due to intermittent nature of distributed generations (DGs), microgrids may not be able to meet the load demand and cause frequency fluctuations. These fluctuations can be migrated using energy storage systems (ESS) to meet the load demand while providing smooth and reliable power. ESS have a wide range of performance requirements in terms of lifetime and discharge capacities. ESS acts as a source of flexibility by providing solutions to several applications like peak shaving, renewable integration, and system reliability. The microgrid configuration with ESS can address a variety of applications such as load leveling, renewable matching, and meeting end user demand by distributing power to remote and isolated localities, to mitigate load fluctuations and improve power quality.^[1] Battery energy storage systems (BESS) are one of the best choices to support the DGs in meeting reliability, as they are the sources of high power and energy densities. The most promising features in terms of performance requirements of the ESS are lifespan and capacity discharge. Most accepted batteries used in grid connected applications are lithium-ion batteries due to their impressive energy density and lifetime. Storage with Li-ion battery is the best option at end-user

application to meet the load criteria while obtaining minimized electricity cost due to its high depth of discharge (DoD).

In the current scenario, microgrid operations with renewable energy sources and BESS are the preferred solutions in the grid and island connected mode to overcome energy crisis.^[2] The main purpose of BESS with the integration of DGs is to obtain optimal power with minimum cost of microgrid operation. The other feature of BESS is that it helps in eliminating the surplus of electricity cost expenses from the grid.^[3] The intermittent nature of DGs can be overcome using a suitable ESS, particularly BESS, due to its high energy and power density. By obtaining optimized maximum discharge rates of BESS within the limits of state of

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charge (SoC), the efficiency of the microgrid can be achieved without any limitations in power supply at the end user level.^[4]

Microgrids aim at obtaining the optimal operational cost incorporating DGs and ESS, preferably battery energy storage. The other objective of a microgrid system is to provide uninterrupted power from DGs and BESS to the customer with minimized cost of electricity (COE). This can be achieved by determining the optimal size of BESS and by optimizing the operational costs related to DGs and BESS. Size of BESS and its lifetime decides the energy efficiency and reliability of the system to achieve the objective of optimal operating cost.^[5] In addition to lifetime of BESS, it is necessary to consider the hourly throughput-based degradation cost with respect to maximum capacity of BESS. The current research problem is defined as a cost benefit analysis of the microgrid by minimizing electricity cost to reduce the customer tariff and minimizing OC as a benefit to the customer considering DGs' power dispatch schedule with OC and BESS degradation parameters. These objectives can be achieved by considering battery degradation effect in determining the OC analysis of BESS.

Many techniques have been applied in literature to minimize the operating and electricity cost by optimizing the size of BESS in terms of its discharge capacities without considering the degradation effect-based lifetime.^[6] Mesh-adaptive direct search algorithm (MADS) is proposed to determine the optimal operating strategy along with OC minimization of microgrid. Adaptive modified particle swarm optimization is proposed in ref. [7] to optimize the OC of microgrids consisting of DGs and BESS over a period of 24 h time horizon.^[8] MADS is proposed to determine the optimal operating strategy of microgrids with OC minimization. In ref. [9], fuzzy logic-based controlled storage is proposed to determine optimal scheduling of microgrids for OC reduction. Large penetration of intermittent wind and solar power sources without suitable storage may weaken the system stability due to their unpredictable nature. Therefore, integrating DGs with energy storage reduces the variable nature of the system and improves the reliability in ref. [10]. Model predictive control (MPC)-based optimization is presented to maximize the total profit of wind power by determining the optimal size of BESS. Adaptive modified firefly algorithm has been proposed in ref. [11] to determine the operation management-based uncertainty of DGs and energy storage devices.^[12] Particle swarm optimization (PSO) is proposed to find the size of BESS while achieving maximum benefit by obtaining minimum OC by means of peak leveling and energy-saving techniques.

The optimal size of BESS is determined in ref. [13] while minimizing the total OC of microgrids considering unit commitment problem using genetic algorithm. In ref. [14], novel scheduling-based algorithm is proposed to optimize the total degradation cost and fuel consumption cost of microgrids in grid-connected mode. FA proposed in ref. [15] minimizes the total OC by considering the capacity-based lifetime degradation model obtaining the size of BESS. In ref. [16], mixed integer linear programming-based optimization has been proposed to determine the lithium-ion battery degradation assessments in microgrids considering SoC range degradation models of BESS. In ref. [17], MPC approach is implemented to integrate the DGs into isolated smart microgrid systems to improve the reliability of the system with BESS, neglecting the degradation effect.^[18] The microgrid OC is minimized by optimizing the size of BESS with

improved bat algorithm by considering the fixed lifetime of BESS.^[19] Probabilistic operation and energy management of DERs in a microgrid system under uncertain environment are presented using adaptive gravitational search algorithm and point estimate method without considering degradation effect of BESS.

From literature, it is observed that few research works were carried out on microgrid OC optimization considering optimal size of BESS and degradation effect. As the size of BESS depends on discharge powers and degradations, determining BESS size and degradation parameters is necessary in obtaining minimum operating and electricity costs. The battery lifetime and degradation cost based on BESS throughputs or discharge are essential factors in calculating the overall system cost, which controls the operating and electricity cost. In the literature, lifetime is predicted to calculate the total cost of BESS by neglecting the throughput-based degradation cost. Few works are carried out to determine the size of BESS considering that degradation effect was implemented based on the capacity-based degradation model. In this model, the battery lifetime is considered based on aging by determining the state of health of the battery and by degradation capacity calculation based on percent degradation and number of cycles specified by the manufacturer. The other models considered in literature to carry out degradation are based on cycle life and calendar life of BESS. In cycle life model, battery aging is determined based on DoD and number of cycles, whereas in calendar life mode, the lifetime of battery is determined based on operating temperatures or SoC. For these two models, the data sheet will be provided by the manufacture, so based on the application the battery can be chosen. However, for operating the microgrid with minimum cost and to provide reliable power, throughput model-based degradation is accurate in obtaining the optimal results of operating and electricity costs. This model determines the hourly degradation cost based on discharge of BESS within the calculate lifetime, so the microgrid operator can accurately know the average OC of the system within BESS lifetime. The main aim of this article is to formulate the total cost of BESS considering throughput-based degradation cost at each time interval within the determined lifetime. Hence, the optimal degradation cost corresponds to minimum average OC within an obtained lifetime with BESS. This work considers hourly predicted DG output powers within its specified capacities by neglecting the randomness and optimized for BESS size in between minimum and maximum capacities by calculating battery degradation cost at each interval and lifetime.

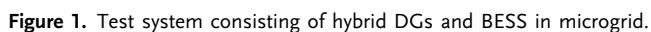
This article presents a hybrid renewable energy system-based microgrid to optimize the OC and electricity cost as objectives considering BESS degradation parameters based on throughput degradation model. The BESS degradation parameters of average kWh throughput, total actual MWh, lifetime, and unit degradation cost are determined based on corresponding power dispatch from DGs and BESS. These obtained BESS degradation parameters are incorporated in the total cost formulation of BESS to minimize the operating and electricity cost. Cost of power dispatch is determined utilizing data of predicted powers from wind PV and microturbines for 24 h. By modeling the BESS and determining the throughputs for 24 h, the objective function is formulated for microgrid OC and COE in grid-connected mode by incorporating the degradation parameters. As the size of BESS depends on the discharge rates, it is necessary to model the

The article is organized as follows. Section 2 describes the system microgrid configuration and components modeling. Section 3 details problem formulation for microgrid OC and COE minimization with BESS degradation simulation parameters. Section 4 presents implementation strategies of objectives with PSO, APSO, JAYA, and LP-IP algorithms. Section 5 elaborates on results and discussions with comparative analysis of OC, electricity cost, degradation costs with average throughputs, and BESS simulation parameters. Section 6 provides conclusions and future scope of the work. Section 7 details the necessary studies that carry out this work.

Figure 1 shows a grid-tied residential-level energy management system with integration of HRES photovoltaic (PVs) wind Turbines (WTs), micro Turbine (MT), and BESS. The inverter used in grid-connected mode is used for power conversion and to deliver the stored energy from battery to the utility grid. The battery is discharged depending on the minimum and maximum energy capacity limits of BESS. The OC and electricity cost of the considered system is optimized by satisfying the load demand considering battery lifetime, while accounting for degradation effect in grid-connected mode. To ensure the continuity of supply in the absence of grid supply due to any outage or etc., microgrid switches in island-connected mode which may be either a manual islanded mode or automatic islanded mode depending on the type and nature of the load.

Due to the above inherent technical limitations, microgrid operation in islanded mode is typically used for disaster recovery situations rather than business continuity services with the use of BESS. In an island connected mode, microgrids provide on-site power generation that supports facility operations indefinitely until the continuity of utility services is restored. Once services are restored, microgrid ensures the continuity of supply. Therefore, in islanded connected mode, depending on the loads and its requirement by proper sizing, the reliability of power supply is ensured. So, irrespective of load and disaster situations, microgrids can be operated in grids connected to ensure more reliable power with integration of HRES and BESS.

A photovoltaic system or solar photovoltaic power is designed to supply required electrical power by converting incident solar radiation to electricity. Therefore, the output power of the solar PV system is a function of solar irradiance which is uncertain. As irradiance is uncertain in nature, the solar output power is also stochastic in nature. Therefore, for modeling of the solar PV system, standard test conditions and normal operating cell temperature of the PV system are considered with solar irradiance operating points.^[20] Maximum power point tracking (MPPT) is equipped with PV system for obtaining the maximum power output and optimizes power between the solar panel and battery



source. The total power generated by PV panels at time ' t ' is given by Equation (1). The solar PV output powers are generated randomly by considering the following equations for different operating points.

$$P_{PV} = P_{STC} \cdot (G_C / G_{STC}) [1 + k(T_C - T_{STC})] \quad (1)$$

where G_C is solar irradiance operating point in (kW m^{-2}), G_{STC} is solar irradiance at STC, P_{STC} is rated power output by PV modules at STC, T_C is cell temperature, T_{STC} is the temperature at STC, and k is the power temperature coefficient.

2.2. Modeling of Wind Power System

Wind power is the process of generating the electricity by converting kinetic energy to mechanical power and thereby converting to electric power through a generator. As wind power output from wind turbines is a function of wind speed, which is stochastic in nature, the wind power generation is also an uncertain variable. As the wind power output is directly proportional to the cube of the wind speed, increased amount of energy can be produced with small changes in wind speed. By considering the uncertainty, output of wind power is expressed in terms of wind speed using wind speed coefficients,^[21] which is a function of rated, cut-in, and cutout wind speed, expressed by Equation (2).

$$\begin{aligned} a + bV_w V_{ci} &\leq V_w \leq V_r \\ P_{WT} &= 0 \quad V_w \leq V_{ci}, \quad V_w \leq V_{co} \\ P_r V_r &\leq V_w \leq V_{co} \end{aligned} \quad (2)$$

where $a = (P_r \cdot V_{ci} / V_{ci} - V_r)$ and $b = (P_r / V_r - V_{ci})$

V_{ci} , V_{co} , V_w , and V_r are the cut in, cutout, nominal, and rated wind speeds, respectively. The behavior of wind speed can be simulated using Weibull distribution function.

2.3. Modeling of Microturbine

Microturbines are single-stage combustion turbines which generate power ranging from few kW to MW. Microturbines are simple forms of gas turbines, featuring turbine rotors and compressors which are powered with natural gas or diesel generators and used for small-scale power applications.^[22] Considering the fuel and installation cost of microturbines, the output powers are generated randomly between minimum to maximum powers and cost function of microturbine can be expressed as in (3), where a and b are cost coefficients taken as 0.0325 and 0.014 and P is output power with respect to maximum power generation from the microturbine.^[23]

$$C(P_{MT}) = a + bP \quad (3)$$

2.4. Modeling of Lithium-Ion Battery Energy Storage Systems

BESS plays a vital role in the microgrid system and can help to solve the intermittent nature of DGs. The most important characteristic of BESS is that it provides emergency backup power to meet the load demand in support of DGs. This improves

reliability and reduces electricity price surge from customer side. Recently, lithium-ion batteries were used in microgrids because of their high energy density and lifetime. In this article, LI-BESS is modeled based on battery degradation cost and lifetime, considering discharge throughputs at each instant of time. The charge and discharge rates at each duration^[24] of BESS are expressed as Equations (4)–(7). BESS discharges its power during each hour, providing support to DGs to meet the load demand and to get optimal microgrid OC and electricity cost.

Charging mode:

$$C_{BES, t+1} = \min\{C_{BES, t} - tP_{BES, t}, C_{BES \min}\} \quad (4)$$

$$P_{BES \text{ charge}(t)} = \max\left\{P_{BES \min}, (C_{BES, t} - C_{BES, \max}) \frac{\text{charge}}{t}\right\} \quad (5)$$

Discharging mode:

$$C_{BES, t+1} = \max\left\{\left(\frac{C_{BES, t} - tP_{BES, t}}{d}\right), C_{BES \min}\right\} \quad (6)$$

$$P_{BES \text{ discharge}(t)} = \min\left\{P_{BES \max}, (C_{BES, t} - C_{BES \min}) \frac{\text{discharge}}{t}\right\} \quad (7)$$

$P_{BES \text{ charge}}$ and $P_{BES \text{ discharge}}$ are the charge and discharge powers at each hour instant. $P_{BES \min}$ and $P_{BES \max}$ are minimum and maximum BESS capacity limits in terms of power. $C_{BES \min}$ and $C_{BES \max}$ are the minimum and maximum stored energy capacity limits of BESS. η_{charge} and $\eta_{\text{discharge}}$ are the charge and discharge efficiencies, which play a significant role in delivering discharging power to the load.

2.5. Utility Grid

The utility grid contains predicted load demand for 24 h, which is to be satisfied by power allocations of DGs and BESS. As the system is in grid-connected mode, the balance of supply that is to be satisfied to meet load demand is taken from the grid. To get the accurate power allocations of DGs and BESS, the hourly predicted load for 24 h is considered. Size of BESS is determined for each hour, which reflects and improves the reliability of load demand. Microgrid OC has been minimized while considering predicted load demand and dispatching required power to the loads.

3. Problem Formulation: Microgrid OC with BESS Degradation Cost and Electricity Cost

In the present work, HRES-based microgrids having 2 units of wind, 2 units of PV, 1 unit of MT, and BESS are considered to meet the hourly predicted load demand to ensure reliability in supply and reduce the intake of power supply from the grid, thereby reducing the electricity cost to customers. The proposed microgrid should satisfy the average load demand of around 1.75 kW by obtaining power from DGs and BESS, considering the degradation effect. This concept of average load indicates

the mean load level to be met by the corresponding DGs and BESS through sizing. This helps to step up the microgrid of capacity 1.75 kW by selecting the appropriate DGs and BESS size. The best way to utilize the renewable sources is the use of HRES. The other major advantages of HRES are continuous power supply, load management, high efficiency, and low maintenance cost as they do not need to require frequent servicing. With these major advantages, HRES are becoming popular and are being installed, wherever a small kW microgrid is required like offices and domestic loads. The objective function is to minimize the OC of microgrid with the use of HRES and correspondingly minimize the electricity cost of customers in grid-connected mode. The microgrid OC comprising operation and maintenance cost of DGs and BESS, total day cost (TDC), and degradation cost of the BESS are given by following equations.

$$\text{Min}(O.C)_{MG} = \sum_{t=1}^T \{D.G_{\text{related cost}} + \text{grid}_{\text{related cost}} + (T.C)_{\text{BESS-DEG}}\} \quad (8)$$

$$\text{Min}(O.C)_{MG} = \text{Min} \sum_{t=1}^T (C_{DG} + MC_{DG} + C_{BESS} + C_{\text{grid}} + \text{DEG}_{BESS} + \text{TDC}_{BESS}) \quad (9)$$

Equations (8) and (9) describe the objective function of microgrid OC considering DG and grid-related cost with total BESS degradation cost. C_{DG} and C_{BESS} are the OCs required to deliver power from respective DGs and BESS. MC_{DG} is the maintenance cost of DGs and TDC_{BESS} is the TDC of BESS, which includes fixed cost (FC), maintenance cost (MC), replacement cost (RC), and rate of returns. C_{grid} gives the cost of grid power required to meet demand. DEG_{BESS} refers to total degradation effect cost in rupees, which is given by the following equation. The degradation effect calculations of BESS parameters to determine degradation cost and lifetime,^[25] which is to be incorporated in microgrid OC problem formulation, are given by the following Equations (10)–(12).

$$\text{DEG}_{BESS} = [W * C_{BESS-\text{Max}}] \quad (10)$$

Here 'W' gives the degradation cost of BESS in Rs kWh⁻¹, which is based on average kWh throughput and to be obtained from simulations and initial cost of BESS, which is given by

$$\text{Battery DEG - Cost (W)} = \left\{ \frac{\text{intial investment cost of battery (Rs)}}{\text{Average kWh throughput of battery}} \right\} \quad (11)$$

The degradation-based lifetime of BESS, which is to be incorporated in TDC, is given by Equation (11). The lifetime (LT) here is obtained based on actual kWh throughput to be obtained from simulations and average lifetime kWh, which is specified by the manufacturer, is given by the following equation.

$$\text{Battery}_{\text{Life Time}}(\text{Years}) = \left\{ \frac{\text{Total actual MWh throughput for year}}{\text{Total average life time MWh}} \right\} \quad (12)$$

The total day cost (TDC) of BESS is given by Equation (13). The final total cost of BESS based on degradation is given by Equation (14).

$$(\text{TDC})_{BESS} = \frac{C_{BESS-\text{Max}}}{365} \left\{ \frac{IR(1+IR)^{LT}}{(1+IR)^{LT}-1} * FC + (RC * MC) \right\} \quad (13)$$

$$(T.C)_{BESS-DEG} = (W * C_{BESS-\text{Max}}) * (\text{TDC})_{BESS} \quad (14)$$

The total COE in Rs kWh⁻¹ which is to be minimized satisfying the load demand is given by the following equation. To obtain the electricity cost, the obtained COE is to be multiplied with total load demand at that instant of time.

$$\text{Cost of Electricity (COE)} = \frac{\sum_{t=1}^T (C_{PV} + C_{BESS} + C_{MT} + \text{DEG}_{BESS})}{\sum_{t=1}^T P_L} \quad (15)$$

The maintenance cost of DGs over a time horizon is given by sum of maintenance cost of individual DGs over total operation time horizon (T) and is expressed as

$$(MC)_{DG} = [(MC)_{WT1} + (MC)_{WT2} + (MC)_{PV1} + (MC)_{PV2} + (MC)_{MT}] * T \quad (16)$$

3.1. Constraints

The microgrid system with HRES and BESS is affected by inequality constraints.

3.1.1. DG's Constraints

$$P_{WT1-\text{Min}} \leq P_{WT1} \leq P_{WT1-\text{Max}} \quad (17)$$

$$P_{WT2-\text{Min}} \leq P_{WT2} \leq P_{WT2-\text{Max}} \quad (18)$$

$$P_{PV1-\text{Min}} \leq P_{PV1} \leq P_{PV1-\text{Max}} \quad (19)$$

$$P_{PV2-\text{Min}} \leq P_{PV2} \leq P_{PV2-\text{Max}} \quad (20)$$

$$P_{MT-\text{Min}} \leq P_{MT} \leq P_{MT-\text{Max}} \quad (21)$$

The above constraints are the maximum and minimum capacity power limits of DGs that have to be satisfied to meet the load demand. The optimal OC and electricity cost obtained have to satisfy the above constraints of DG.

3.1.2. BESS Constraints

$$P_{BESS-\text{Min}} \leq P_{BESS} \leq P_{BESS-\text{Max}} \quad (22)$$

$$\text{SoC}_{\text{Min}} \leq \text{SoC}_{BESS} \leq \text{SoC}_{\text{Max}} \quad (23)$$

The LI-BESS size should be optimized within minimum and maximum power capacities of 50–500 kW with storage capacity of 500 kWh. SoC estimation is a key component for a battery management system which is based on energy storage capability. SoC of LI-BESS should be in between 0 and 1 for each hourly discharge duration which indicates fully discharged and charged, respectively. The reason to maintain SoC of LI-BESS in between 0 and 1 is because of its high energy density, constant power discharge capability, and long life. The typical SoC of LI-BESS is around 80–90%.^[26] The degradation has been obtained by calculating the lifetime of BESS base on hourly discharge maintain SoC range within the limits. If the SoC drops to around 35% of the SoC, then BESS suffers and goes down to age-related loss ready to be replaced within its lifetime if future SoC decreases. Size of BESS in terms of average kWh throughput and actual kWh throughput has to be determined so that the minimum microgrid operating and electricity cost is obtained.

3.1.3. Grid Constraints

The cost of power delivered from the grid is incorporated in microgrid OC problem formulation to improve the accuracy in obtaining the optimal solution. The balance of supply, after obtaining the power allocations from DGs, is taken from the grid to meet the load demand. The balance of supply required to meet the load demand after dispatching power from DGs and BESS is taken from the grid considering the unit prices of the grid.^[27] The maximum and minimum power limits from the grid at any instant of time are given as follows:

$$P_{\text{grid-Min}} \leq P_{\text{grid}} \leq P_{\text{grid-Max}} \quad (24)$$

4. Implementation Strategy with Heuristic and LP-IP Optimization Algorithms

The objective function for OC and electricity bill minimization considering battery degradation cost and lifetime is solved using heuristics optimization and solver-based LP-IP approach. The major advantage of LP-IP-solver based approach is that it provides best optimal solutions to scheduling base cost minimization objectives, in comparison to heuristic approach. Under heuristic optimization techniques, PSO, APSO, and JAYA algorithms are used to solve the objective. LP-IP solver-based approach is used to solve the problem and compared with heuristic techniques to show the effectiveness of LP-IP solver. Results of OC and electricity bill minimizations, degradation costs and lifetime, average kWh, and actual kWh throughputs of microgrid for 24 h have been presented in a comparative analysis.

4.1. Particle Swarm Optimization

PSO is a population-based meta heuristic algorithm proposed by Kennedy and Eberhart in 1995 by simulating the social behavior of birds.^[28] The PSO algorithm starts with the initialization of random particle positions. To prevent particles leaving the search space during first iteration, velocities are initialized to 0. The

steps in algorithm include fitness evaluation, updation of individual best and global best solutions, and correspondingly updating the position and velocities of the particles until stopping criteria is reached. The position and velocity update equations are expressed in (25, 26). The main disadvantage of PSO is acceleration and premature convergence of algorithm, which can be overcome by PSO.

$$V_{i+1} = w * V_i + C_1 * R_1 * (P_{\text{best}} - X_i) + C_2 * R_2 * (G_{\text{best}} - X_i) \quad (25)$$

$$X_{i+1} = X_i + V_{i+1} \quad (26)$$

where V_{i+1} and X_{i+1} are the velocity update and position update equations of each particle in the search space in iteration $i + 1$ and ' i ' is the present iteration. P_{best} and G_{best} are the individual best and global best particle positions of i^{th} iteration, R_1 and R_2 are the random number initializations in between 0 and 1, C_1 and C_2 are learning factors which relate the importance to personal best and global best particles, and ' w ' is the inertia weight constant which starts from positive value and decreases as the iterations progress (this work considers the learning factors C_1 and C_2 as 2 and inertia weight ' w ' in between 0 and 1).

The major advantages of PSO are easy implementation, derivative free, and very efficient global search approach. PSO uses both P_{best} and G_{best} , which is a disadvantage, causing premature convergence. The PSO algorithm is convergent if the velocity of the particles decreases to zero or stays unchanged until the end of iterations. Inertia weight is considered, which is constant, which is similar to introducing virtual mass to stabilize the motion of particles for quick convergence of PSO algorithm. The issue with PSO is premature convergence and weak local search ability. To improve the solution quality of PSO and to improve accuracy in convergence, accelerated PSO (APSO) is implemented.

4.2. Accelerated Particle Swarm Optimization

The APSO is the improved version of standard PSO, which uses only global best (G_{best}) to accelerate the convergence of the algorithm. The standard PSO uses particle best to increase the diversity in the solution quality, which causes low quality in solutions. The convergence of the algorithm can be accelerated using only global best but not both G_{best} and P_{best} . APSO is the solution of all above shortcomings. The diversity in solutions can be improved using some randomness instead of P_{best} given by the equation.^[29] Thus, the velocity update equation in APSO is generated using randomness factor in place of individual best and is given by the following equation:

$$V_{i+1} = V_i + (\alpha * R_1) + R_2 * (G_{\text{best}} - X_i) \quad (27)$$

where R_1 and R_2 are the random numbers between (0, 1). Further improvement in APSO is to accelerate the convergence using a monotonically decreasing function which is given by (28)

$$\alpha = \alpha_0 \gamma^t \quad (28)$$

where α_0 is the initial randomness value which varies between (0.5, 1). In this study, value of α_0 is taken as 0.5 and ' t ' is the total number of iterations. The above equation improves the diversity in solution by reducing randomness. Parameters used

for simulation are learning factors $C_1 = C_2 = 2$, maximum inertia ($W_{\max}$) = 0.9, and minimum inertia ($W_{\min}$) = 0.1; total number of iterations for both the methods is taken as 200.

4.3. JAYA Optimization

Jaya algorithm is a population-based algorithm developed by Rao^[30] to solve various types of unconstrained and constrained optimization problems. Unlike other optimization techniques, JAYA does not have algorithm specific parameters. The only two controlling parameters used in JAYA are population size and total number of iterations. JAYA algorithm considers best and worst values of the function and provides optimal results in less number of iterations. JAYA algorithm achieves successes by providing the best optimal solutions by moving away from the worst solution. The position update equation of JAYA algorithm is given by Equation (29), considering the worst value of function to improve the quality of solution.

$$X'_{ij} = X_{ij} + R_1(G_{\text{best},j} - |X_{ij}|) - R_2(G_{\text{worst},j} - |X_{ij}|) \quad (29)$$

$G_{\text{best},j}$ and $G_{\text{worst},j}$ are the best and worst solutions of the j^{th} variable respectively. X'_{ij} is the position update vector, $|X_{ij}|$ is the absolute value of the current position, and R_1, R_2 are the random numbers between (0,1). The JAYA algorithm always moves closer to success, avoiding the worst solutions. The other main advantage of JAYA is that it always moves closer to success to avoid the infeasible solutions.

The premature convergence is avoided by JAYA by considering the worst solution in the position update equation; thereby, the tendency to avoid the worst solution is improved and optimal solution tends to approach global best solution. The issue with JAYA algorithm is having slow convergence. As optimal solution tends to move away from the worst solution, there are some variables in some dimensions that are close to optimal solutions which are considered to be avoided, as this JAYA algorithm suffers with slow convergence even though it gives optimized results in less iterations.

4.4. Linear Programming-Based Interior Point (LP-IP) Solver

LP-IP solver is used to solve both linear and nonlinear optimization problems considering inequality and equality constraints as variables. LP-IP solver finds minimum of a function specified by the following expression.

$$\min_x f(x) = \begin{cases} A \cdot x \leq b \\ A_{\text{eq}} \cdot x = b_{\text{eq}} \\ lb \leq x \leq ub \end{cases} \quad (30)$$

where $f(x)$, b , b_{eq} , lb , ub are vectors and A and A_{eq} are matrices. A and b correspond to inequality constraints. A_{eq} and b_{eq} correspond to equality constraints and the lower bounds (lb) and upper bounds (ub) correspond to limits of the function in which the objective is to be optimized. LP-IP solver gives optimal results because the objective function itself acts as a barrier which penalizes nonfeasible solutions satisfying the inequalities $x \geq 0$ using Karush–Khun–Tucker (KKT) conditions (higher-order derivatives

of functions),^[31] which helps in obtaining the minimized optimal solutions within the selected value of function tolerance. LP-IP method can handle the objective functions that are discontinuous by approximating the constraints of objectives as a set of boundaries by modifying them without reaching global minima. The main advantage of LP-IP solver is that it helps in re-evaluation of a basic plan for varying conditions. Therefore, LP-IP solver algorithm is more accurate in solving the constrained minimization of the function within the bounds and converges to a global optimal solution.

The LP-IP relies on having a linear programming model with objective function and all the constraints being continuous and twice continuously differentiable. LP-IP solver expresses the objective function with all the constraints, initial values, lower and upper bounds in the form of vectors, and matrices in a 'linprog' function and calls for IP solver to execute the function. The syntax of LP-IP solver is given as in Equation (31) and (32)

$$(X, f_{\text{val}}, \text{exit flag}) = \text{linprog}(\text{fun}, A_{\text{eq}}, b_{\text{eq}}, A, b, lb, ub, \text{options}) \quad (31)$$

$$\text{Options} = \text{optimoptions}(\text{'linprog'}, \text{'Algorithm'}, \text{'interior'}, \text{'point'}, \text{'display'}, \text{'iter'}) \quad (32)$$

As 'linprog' function uses interior point algorithm, the objective function uses the maximum of 200 iterations. "X" indicates the optimized value of the function and f_{val} indicates the corresponding values of objective function, where the optimal result has been obtained. Here, the maximum value of tolerance is 1e-6 which is fixed for LP-IP solver.

4.5. Algorithm of LP-IP Solver-Based approach

The proposed algorithm for microgrid OC and electricity cost minimization considering BESS degradation effect is described in the following steps: 1) Read the input data (DGs' power generation profiles, BESS maximum and minimum capacity limits, and load profile); 2) Initialize population size and cost coefficients of DGs and BESS; 3) Develop and run the DGs and BESS scheduling subalgorithms and check for violation of constraints; 4) Obtain the objective function values of each subalgorithm from step 3 and then initialize assignment matrix with those values; 5) Incorporate the assignment matrix as initializations of objective function values in an optimizer or solver (LP-IP); 6) Check for all constraints like equality, inequality, lower and upper bounds, and objective function matrix allocation in LP-IP solver; 7) If yes, then use the function called "optimoptions" which describes algorithm type and uses "disp" and "iter" options to execute the solver; and 8) Run the program until the max iterations and with tolerance criteria, which indicate the minimized value of the function while satisfying the constraints.

5. Results and Discussions

The microgrid OC and electricity bill minimization considering degradation effect with simulation of various BESS degradation parameters have been studied for 24 h and solved using heuristic and solver-based techniques. Hourly power generation from

wind, PV, and microturbine are considered, having maximum generation limits for a time horizon of 24 h. The objective of the problem is to obtain optimal operating and electricity costs while satisfying the load demand. The above objective is achieved by simulating the BESS degradation parameters such as average and actual kWh throughputs, degradation cost, and lifetime. The throughputs of BESS obtained for each hour help in determining the BESS degradation costs and lifetime to DG output. The BESS degradation simulation parameter has been considered to optimize overall operating and electricity cost and correspondingly average costs are determined.

Table 1 describes the hourly maximum power generation limit of each DG with predicted load demand. The microgrid operating and electricity cost results are obtained, satisfying the optimal power allocation of DGs without exceeding its maximum generation limit. Therefore, the proposed microgrid system should optimize the operating and electricity cost, satisfying the hourly load demand of obtaining power from DGs and BESS considering the degradation effect. The problem is solved using PSO, APSO, JAYA, and LP-IP solver-based methods. Comparison of heuristic approaches and LP-IP solver-based method for operating and electricity costs along with the BESS degradation simulation parameters are presented.

The following sets of optimal results are obtained for heuristic and LP-IP solver-based methods for different cost-related objectives, throughputs, and lifetimes of BESS. The microgrid system

Table 1. Maximum generation limits on DGs and predicted load.

Time [H]	PWT1 [kW]	PWT2 [kW]	PMT [kW]	PV1 [kW]	PV2 [kW]	Load [kW]
1.	660	688	429	0	0	1471
2.	699	707	442	0	0	1325
3.	700	698	367	0	0	1263
4.	666	576	450	0	0	1229
5.	669	675	450	0	0	1321
6.	719	674	351	0	15	1509
7.	711	693	532	10	71	1663
8.	712	732	497	67	90	1657
9.	716	746	504	98	116	1643
10.	706	686	507	122	140	1643
11.	678	661	366	139	155	1652
12.	697	638	372	145	163	1666
13.	700	561	350	145	163	1600
14.	693	650	400	133	155	1642
15.	700	652	600	120	133	1640
16.	697	657	138	94	107	1676
17.	700	660	381	61	86	1920
18.	699	659	617	17	46	2000
19.	644	668	652	0	1	2210
20.	674	664	706	0	0	2220
21.	677	661	744	0	0	2150
22.	688	642	696	0	0	1903
23.	694	674	711	0	0	1666
24.	672	649	711	0	0	1665

is optimized to determine the hourly throughput of the BESS in between its minimum to maximum power limits for 24 h. Comparative analysis has been presented for each hour BESS degradation simulations like throughputs, degradation cost, and lifetime. For corresponding BESS simulations, the operating and electricity cost considering degradation effect has been determined and a comparison is presented to show the effectiveness of LP-IP solver with heuristic methods.

Table 2 describes the lifetime in comparison with the average and actual throughputs from BESS. It is observed that the lifetime of BESS gets improved in proportion with throughput-based discharge from BESS. More the amount of throughputs from BESS, greater will be the lifetime. From the above, solver-based method gives better results in terms of lifetime and throughput.

Table 3 describes the comparative performance analysis of the BESS and microgrid in terms of degradation cost and OC. Degradation cost of BESS is determined from Equation (11), considering investment cost and average kWh throughput, and correspondingly OC of microgrid has been determined from obtained degradation cost. From the above, it is evident that solver-based method gives better performance in terms of system-related costs (**Table 4**).

The table gives the results of microgrid OC considering BESS with and without degradation effect. Equation (9) concludes that the degradation cost effect acts as a burden on the OC problem formulation. Therefore, from the above results, it can be concluded that the OC increases by considering degradation effect but there is gradual improvement in microgrid cost reduction with LP-IP solver when compared to heuristics.

The degradation effect acts as a burden on OC due to which the cost increases, because it is an additional cost on the system which is added to OC. For electricity cost, degradation effect helps in reducing the electricity bill because the throughput of BESS increases due to inclusion of degradation cost, which is more advantageous in minimizing the electricity cost with respect to load demand. This can be concluded from **Table 5** that

Table 2. Lifetime comparisons with average and actual throughputs from BESS.

Algorithm	Average throughput [kWh]	Actual throughput [MWh] for year	Obtained LT years
PSO	393	3.352	3.04
APSO	398	3.482	3.15
JAYA	407	3.565	3.25
LP-IP solver	420	3.704	3.40

Table 3. Average BESS degradation cost comparison with microgrid OC.

Algorithm	Unit degradation cost of BESS [Rs kWh ⁻¹]	OC [Rs] with BESS
PSO	102.89	6501
APSO	100.50	6475
JAYA	98.29	6420
LP-IP Solver	95.20	6201

Table 4. Average microgrid OC comparison with and without BESS degradation effect.

Algorithm	OC with BESS without degradation effect [Rs]	OC with BESS considering degradation effect [Rs]
PSO	6430	6501
APSO	6405	6475
JAYA	6325	6420
LP-IP solver	6160	6201

reduction in average electricity cost when considered with BESS degradation effect. LP-IP solver method gives optimal value of electricity bill minimization and among heuristics JAYA algorithm gives optimal result when compared to other heuristics. Electricity costs vary for 24 h with respect to BESS size and DGs' optimized power meeting the load demand. The electricity cost in this work mainly depends on DGs and BESS size, as they are obtained by optimizing the predictive maximum and minimum powers.

From **Table 6**, it can be concluded that degradation effect plays a major in OC and electricity cost minimizations. Finally JAYA algorithm among heuristics gives optimal results and from all implemented algorithms LP-IP solver method gives better optimal results.

Figure 2 shows the comparison between JAYA and LP-IP solver method and it is clear that the proposed LP-IP solver gives optimal results for OC minimization and delivers best possible solutions for scheduling-based problems. **Figure 3** gives the comparative performance analysis of implemented algorithms for OC minimization. It can be concluded that among performed heuristic-based methods, JAYA optimization algorithm performs

Table 5. Average electricity cost comparison with and without degradation effect.

Algorithm	Average electricity cost without degradation effect [Rs]	Average electricity cost with degradation effect [Rs]
PSO	13 230	13 105
APSO	13 120	12 975
JAYA	12 948	12 840
LP-IP solver	12 844	12 746

Table 6. Average operating and electricity cost comparisons with degradation effect.

Algorithm	Average OC with BESS degradation effect [Rs]	Average electricity cost with BESS degradation effect [Rs]
PSO	6501	13 105
APSO	6475	12 975
JAYA	6420	12 840
LP-IP solver	6201	12 746

better among heuristics in most of the cases. It is observed that LP-IP solver effectively minimizes the objective considering the degradation effect while satisfying the load demand for 24 h among the performed algorithms.

Figure 4 and **5** show the simulation results of power drawn from the grid (error) for 24 h using heuristics and solver-based methods. Error is defined as balance of supply required drawn from the grid in order to meet while obtaining the optimal microgrid OC. From the above results of grid power generations, it can be concluded that solver-based method gives optimal error compared to heuristic approaches because maximum amount of throughputs from BESS has been achieved considering degradation effect. The LP-IP solver-based optimal power discharge from BESS and DGs in satisfying the load demand in grid connected mode is given by the following.

Table 7 shows the power allocations of DG's and BESS in the grid-connected mode, satisfying the load demand. Here P_{GRID} gives the balance of supply that is taken from the grid. Therefore, in order to improve the reliability and reducing the cost of the system, HRES with sizing of BESS plays a vital role in providing required load demand of few kW to MW in remote areas by reducing the burden on the grid.

In another study,^[32] the optimized unit electricity cost of three regions in West Africa has been determined, neglecting battery degradation effect. The average unit electricity cost of three regions has been obtained. We have considered the battery sizing and degradation parameters for above objective function and compared the results of average unit electricity prices of three regions and presented. The forecast data of the performed work have been considered by incorporating battery sizing and degradation parameters and the results obtained with LP-IP solver algorithm are given in **Table 8**.

The main reasons to obtain optimal solutions from LP-IP solver-based methods are that LP-IP solver converts the objective into simplified form of the constraint-based function called "Preslove". Optimizing with generation of initial solutions in the form of vectors based on KKT conditions.^[24] These initial generations of solutions act as a barrier for nonfeasible solutions of the objectives, which is to be incorporated in LP-IP solver and provides optimal solutions within specified value of tolerance. This is the reason that LP-IP solver-based techniques direct the objective in arriving at the promised solutions of optimal power allocations and microgrid OC minimization.

LP-IP method can handle the objective functions that are discontinuous by approximating the constraints of objective as a set of boundaries by modifying them without reaching global minima. The other main advantage of LP-IP solver is that it helps in re-evaluation of a basic plan for varying conditions. Therefore, LP-IP solver algorithm is more accurate in solving the constrained minimization of the function within the bounds and gives optimal solution which is global minima.

Table 9 gives description of simulation parameters for Hybrid DGs and LI- BESS.^[33] The optimal operating and electricity cost is obtained while maintaining maximum and minimum power generation limits of DGs and BESS. Whereas for hourly scheduling of DGs and BESS throughputs, obtained results should not exceed the maximum ratings.

Table 10 gives the simulation parameters of BESS considering degradation effect. Considering above BESS parameters, the

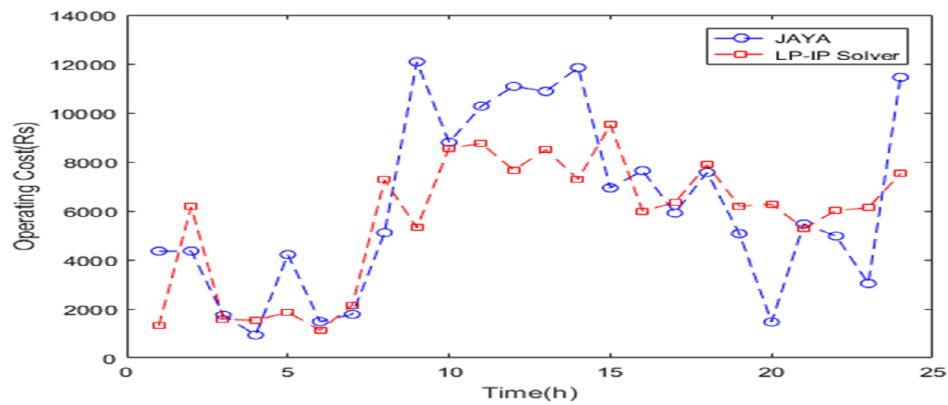


Figure 2. Microgrid OC variation with Jaya algorithm and LP-IP solver.

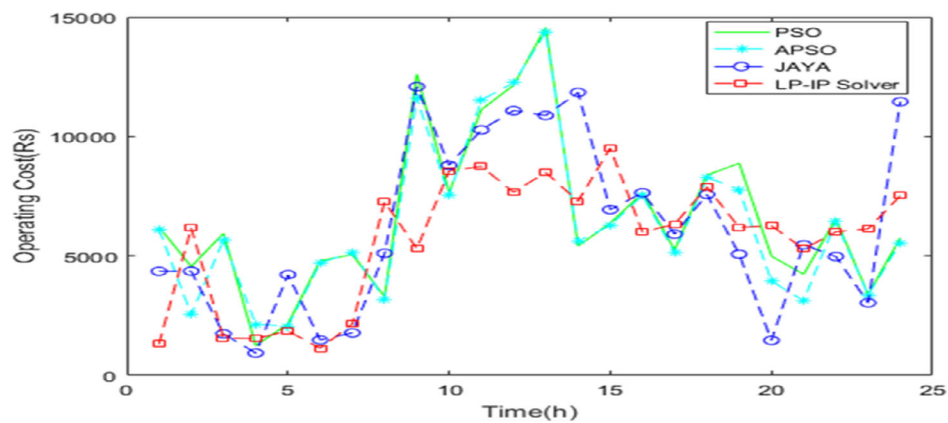


Figure 3. Microgrid OC comparison with heuristics and LP-IP solver methods.

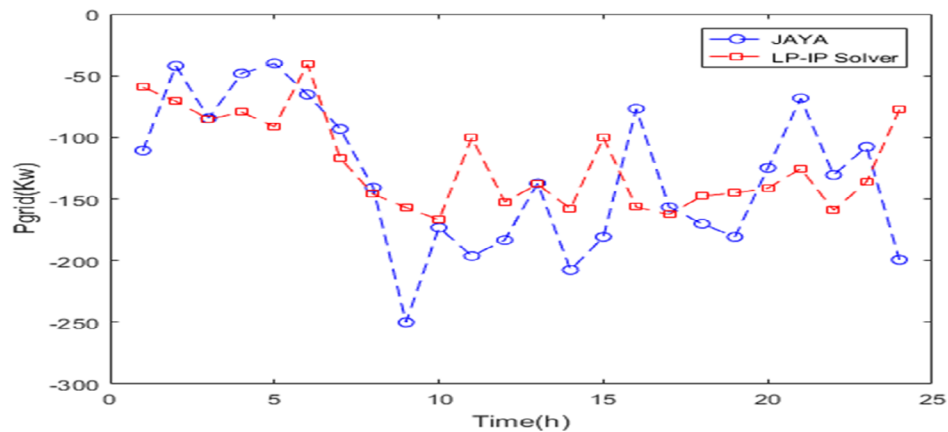


Figure 4. Error plot variation with JAYA and LP-IP solver methods for 24 h.

degradation cost values and lifetimes are determined. Degradation cost and lifetime of BESS can be obtained by knowing the battery investment cost and total average kWh throughput. Using the degradation cost parameters, the OC and electricity cost is to be optimized. Therefore, BESS degradation plays an important role in improving the accuracy of the microgrid system with DERs.

6. Conclusions and Future Scope

In this article, OC of microgrid has been minimized, considering battery degradation effect-based lifetime and correspondingly determined electricity cost to customers considering hybrid renewable energy resources. The objective is solved by determining the BESS degradation parameters and implementing with

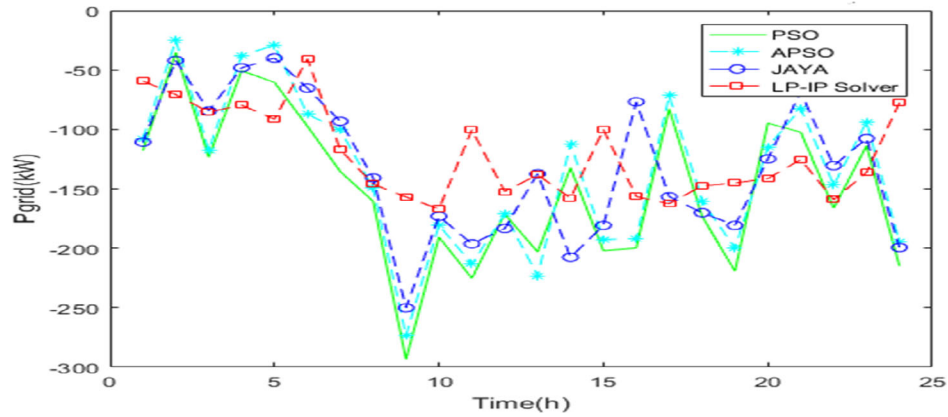


Figure 5. Error plot with comparison heuristics and solver methods for 24 h.

Table 7. Optimal power discharges from DGs and BESS in grid-connected mode.

Time [H]	PMT1 [kW]	PMT2 [kW]	PWT [kW]	PV1 [kW]	PV2 [kW]	PBESS [kW]	PGRID [kW]
1	483.90	246.45	258.63	0	0	428.75	−55.3
2	499.33	571.29	207.88	0	0	46.27	−65.8
3	486.63	622.67	194.40	0	0	128.21	−83.5
4	74.83	576.46	114.15	0	0	407.11	−81.5
5	296.66	250.26	290.46	0	0	306.27	−86.4
6	537.41	296.11	217.88	0	0	275.19	−37.8
7	417.49	490.64	462.92	0	4.8277	485.26	−115
8	519.67	546.17	360.11	9.79	21.27	324.10	−142.1
9	477.90	430.64	433.24	42.11	16.52	304.36	−182.9
10	409.78	478.83	377.06	32.280	108.51	366.98	−175
11	306.86	352.27	269.38	107.53	77.35	431.18	−98.4
12	429.18	550.53	368.87	115.36	100.392	90.68	−148
13	454.20	389.92	324.91	149.908	108.291	473.41	−135.2
14	482.91	426.47	408.55	94.247	156.679	220.76	−155.9
15	367.85	394.05	361.57	104.128	114.756	343.44	−98.1
16	371.25	511.54	513.15	25.627	90.217	300.44	−154.6
17	255.92	490.26	468.40	87.093	36.553	260.28	−160.6
18	427.18	504.80	463.00	46.272	1.383	445.27	−144.6
19	546.36	474.49	547.31	4.679	2.503	300.43	−143.1
20	539.55	405.20	541.59	0	0.7078	474.52	−139.4
21	550.90	595.52	558.45	0	0	445.27	−122.4
22	496.24	515.35	465.65	0	0	476.68	−155.2
23	410.64	557.96	598.63	0	0	402.42	−133.2
24	522.00	426.40	439.39	0	0	418.26	−74.5

heuristic and LP-IP solver-based techniques. The average and actual throughputs have been obtained from BESS parameters, by which the degradation cost and lifetime is determined. The results show that the LP-IP solver-based approach gives optimal results for minimized microgrid OC considering BESS degradation effect and its simulation parameters, satisfying the load demand for 24 h. The effect of degradation cost minimizes the electricity cost and slightly increases the OC when compared

without degradation effect. Even though there is increase in OC, it will be beneficial to the microgrid operator for avoiding future expenses, so that accurate BESS size can be selected for a microgrid based on application and reduction in electricity cost benefits to the electricity consumer. For battery capacity of 500 kWh, the optimal size of battery is determined to be in the range of (350–450 kWh) for which the OC obtained is minimum. The above specified range provides the minimized OC and electricity

Table 8. Unit electricity cost comparisons with and without BESS degradation and sizing.

Regions in West Africa	Average unit electricity cost [\$/kWh] neglecting battery degradation parameters and sizing	Average unit electricity cost [\$/kWh] considering battery degradation parameters and sizing
Navrongo	0.321	0.309
Kumasi	0.302	0.285
Accra	0.287	0.275

Table 9. Simulation parameters for DGs and BESS.

Simulation parameters	Values
Population size	40
Prediction horizon	24 h
Maximum number of iterations	200
OC of PV	67 Rs kWh ⁻¹
OC of wind	60 Rs kWh ⁻¹
OC of MT	49 Rs kWh ⁻¹
BESS power capacities	50–500 kW
Wind turbine ratings	0–700 kW
Microturbine ratings	0–700 kW
Solar PV ratings	0–150 kW
Grid power ratings	0–300 kW

Table 10. Simulation parameters for BESS with degradation effect.

BESS parameters with degradation effect	Values
Battery capacity	500 kWh
Battery power ratings	50–500 kW
Battery investment cost	45 000 Rs
Maintenance cost	500 Rs kWh ⁻¹
Replacement cost	65 Rs kWh ⁻¹
Total average life time specified for one year	1.1 MWh
Interest rate	0.06

cost with BESS, so that load demand is met by taking reduced amount of power intake from the grid. Based on simulation results, proposed LP-IP solver algorithm gives optimal results of battery degradation, which is found to be 95.20 Rs kWh⁻¹ for which the average microgrid OC is found to be optimum. It is concluded from the results, that the OC of the system is high for smaller BESS size and reduces for large BESS size. This work can be extended by considering the capacity-based degradation depending on percentage degradation and SOC range to optimize the operating and electricity cost considering HESS.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

battery energy storage systems, degradation effects, distributed generations, microgrids, optimizations, throughputs

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